

Amodal Instance Segmentation with IRAIS Dataset for Sim-to-Real Transfer

Bidong Chen^{† 1 2} Lingui Li^{† 3}

Abstract

Amodal instance segmentation is hindered by the scarcity of scalable and transferable annotations. We introduce MaviGen, an automated 3D retail scene modeling and rendering framework that generates photorealistic multi-view images with complete amodal masks. Building on MaviGen, we present the IRAIS dataset, a sim-to-real benchmark comprising a large-scale synthetic multi-view set (3D-IRAIS) and a human-annotated real image set (Real-IRAIS), both sharing unified label definitions and evaluation protocols to facilitate rigorous transfer studies. We propose EUREKA, an encoder-only, query-efficient network for amodal instance segmentation that performs full-image multi-task inference via unified amodal/visible queries and dual mask heads. The dual heads enable mutual supervision between complete and visible masks, while the conditional masked self-attention mechanism further strengthens occlusion reasoning. Experiments establish strong baselines on IRAIS and achieve state-of-the-art performance on D2SA and COCOA-cls, demonstrating substantial improvements in sim-to-real transfer.

1. Introduction

Amodal Instance Segmentation (AIS) aims to recover complete masks, including occluded regions, thus enabling robust scene understanding. This capability is particularly critical in retail, where densely packed products are frequently occluded by carriers and fixtures (e.g., carts, baskets, and



Figure 1. Qualitative results of our model on four AIS benchmarks: 3D-IRAIS, Real-IRAIS, D2SA, and COCOA-cls. Each example overlays the predicted amodal instance masks on the RGB image.

shelves), making amodal perception essential for inventory analysis and automated checkout (Chen et al., 2022).

The advancement of AIS is closely linked to the development and quality of datasets. Amodal annotation on real-world images is inherently ambiguous and expensive: annotators must infer invisible shapes, which often results in noisy and inconsistent labels and limits scalability (Ao et al., 2024). Meanwhile, widely used AIS benchmarks such as COCOA (Zhu et al., 2017), KINS (Qi et al., 2019) are not tailored to real-world retail. While D2SA (Follmann et al., 2018) is limited to a tabletop setup with only a few product categories; overall, these datasets mainly cover generic objects and scenes and typically lack geometry-aware supervision (e.g., depth, occlusion order, and camera parameters), which is crucial for controlled occlusion analysis and sim-to-real evaluation. Retail datasets such as ORFD (Geng et al., 2018) are valuable for recognition but do not provide amodal masks, and many retail datasets focus on visible-only recognition with limited category coverage and carrier diversity. As also noted in recent work (Tai et al., 2025), existing amodal datasets remain constrained in scale, domain coverage, and metadata richness, leaving retail AIS without a large-scale benchmark with comprehensive cross-modal signals that supports rigorous sim-to-real transfer.

On the modeling side, AIS methods have evolved from CNN-based occlusion reasoning (e.g., Mask R-CNN (He et al., 2017), OR-CNN (Follmann et al., 2019), BCNet (Ke

[†] Corresponding author. ¹Department of Informatics Engineering, University of Coimbra, Coimbra, Portugal. ²Faculty of Applied Sciences, Macao Polytechnic University, Macao, China. ³School of Modern Information Industry, Guangzhou College of Commerce, Guangzhou, China.. Correspondence to: Bidong Chen <bidong@dei.uc.pt, bidong.chen@mpu.edu.mo, cbdchen88@gmail.com>, Lingui Li <202212090060@xs.gcc.edu.cn>.

et al., 2021), ASPNet (Mohan & Valada, 2022), VRSP-Net (Xiao et al., 2021), ASN (Qi et al., 2019)). More recently, they have progressed to transformer-based global modeling (e.g., AISFormer (Tran et al., 2022), Shapeformer (Tran et al., 2024), and Mask2Former (Zhang et al., 2022)), together with complementary advances in boundary and shape modeling and generalization (e.g., AIFormer+SAMEO and RTMDet+SAMEO (Tai et al., 2025), Chen et al.’s model (Chen et al., 2025)).

Despite this progress, retail deployment remains challenging. First, severe occlusions from dense stacking weaken available visual cues, causing models to overfit to partial textures and generate implausible completions. Second, retail imagery exhibits domain-specific characteristics, including specular packaging, repetitive layouts, and frequent truncation by shelves, violating shape priors learned from generic benchmarks. Third, existing data generation methodologies are often insufficiently specified, and the lack of geometry-aware supervision (e.g., depth, occlusion order, and camera parameters) results in substantial sim-to-real transfer degradation.

To address these challenges, we adopt a dataset-model co-design strategy. We propose **MaviGen**, a detailed methodology for 3D retail scene generation and rendering with controllable multi-view acquisition and geometry-aware supervision, including amodal and visible masks, depth maps, occlusion order, and camera parameters. Based on MaviGen, we introduce IRAIS, a retail sim-to-real benchmark that comprises a large-scale synthetic split (3D-IRAIS, generated with MaviGen) and a human-annotated real split (Real-IRAIS). The two splits share the same label names and definitions, and the annotation guidelines follow COCOA, enabling controlled sim-to-real transfer studies under a consistent evaluation protocol. We further propose **EUREKA**, an *streamlined encoder-only ViT* architecture for joint amodal and visible learning. Paired amodal and visible supervision plays a critical role in enabling EUREKA to achieve robust performance under severe occlusion and bridging the gap between simulation and real-world (sim-to-real) application. Our contributions are as follows:

- **MaviGen**: A comprehensive 3D retail scene generation and rendering pipeline that supports scalable multi-view synthesis and provides geometry-aware signals, including depth information, occlusion ordering, and camera parameters.
- **IRAIS**: A sim-to-real amodal instance segmentation benchmark in retail scenarios, comprising 3D-IRAIS (scalable synthetic multi-view rich source data) and Real-IRAIS (high-quality few-shot real-world test set).
- **EUREKA**: We propose a streamlined encoder-only ViT segmentation model that combines single-modality

RGB image input, unified queries, and an improved conditional masked attention mechanism to achieve cross-modal instance segmentation. The model achieves state-of-the-art results on the 3D-IRAIS, Real-IRAIS, COCOA-cls, and D2SA datasets, and validates the feasibility of sim-to-real transfer via IRAIS.

2. Proposed Pipeline

2.1. 3D Model Building

3D Model Acquisition and Reconstruction. As shown in Figure 2 (a), we establish a standardized pipeline for building 3D product models that combines high-fidelity geometry acquisition with appearance modeling, with all outputs further refined by 3D artists. We provide two complementary routes to obtain the underlying geometry. (1) *3D scanning*: we use a *Skyworth 3D Otter Lite* scanner with a motorized turntable for omnidirectional capture. A light-absorbing panel reduces ambient interference, and a silver reflector improves illumination uniformity, enabling robust acquisition of an untextured mesh with high geometric fidelity. (2) *2D image-based multi-view reconstruction*: we capture images from six canonical viewpoints (up, down, front, back, left, and right), from which artists reconstruct and refine the geometry. For all products with colored appearances, we adopt a unified workflow: after obtaining a clean digital 3D geometry model from either route, we apply 2D image-based texture mapping to obtain the final surface appearance. During texturing, material properties (e.g., gloss, roughness, and color) are considered to better differentiate visually similar products with distinct surface appearances. This pipeline captures both structural and material cues, providing reliable 3D assets for downstream data generation.

2.2. MaviGen Pipeline

Object Assets and Carrier Analysis. As shown in Figure 2 (b), MaviGen organizes all objects and carriers into a unified asset library with predefined mass and collision properties, supporting both physics-based and geometry-constrained placement. To enhance diversity, objects are randomly sampled with replacement, retaining their original poses and meshes. Carriers (from simple planes to complex containers) are randomly initialized or user-specified to serve as anchors for placement and for hemispherical viewpoint sampling.

Create the Placement Plane and Viewpoints Sampling. As shown in Figure 2 (b), given a carrier C with vertex set V_c , its axis-aligned bounding box is defined as

$$\min_c = (\min v_x, \min v_y, \min v_z) \quad (1)$$

$$\max_c = (\max v_x, \max v_y, \max v_z) \quad (2)$$

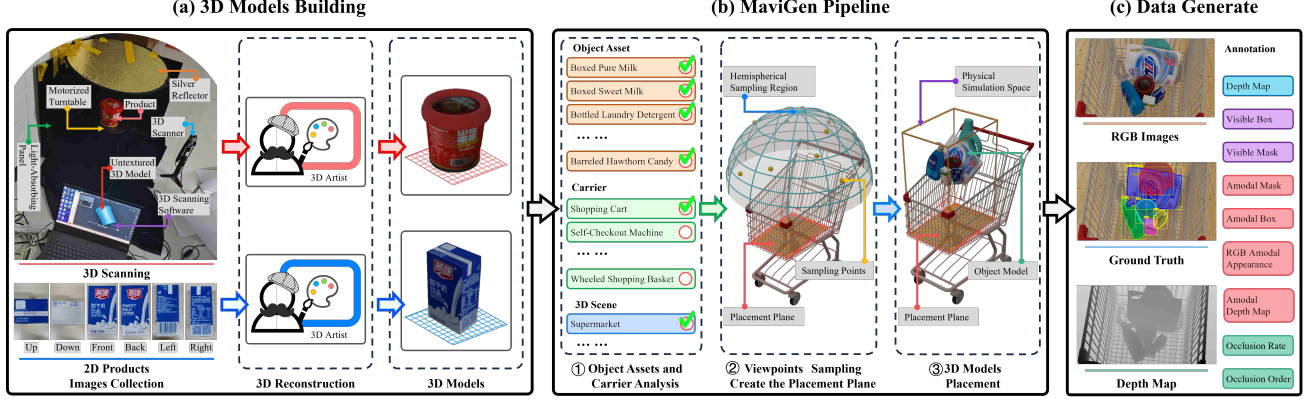


Figure 2. Overview of the **MaviGen** data generation pipeline. It consists of three stages: (a) **3D Model Building**, where product models are created via **3D** scanning or multi-view **2D**-based **3D** reconstruction; (b) **MaviGen Pipeline**, which analyzes object assets and carriers, samples viewpoints and placement planes, and places **3D** models in a physically simulated environment; (c) **Data Generation**, which produces multi-modal outputs including RGB images, depth maps, visible/amodal masks and boxes, RGB amodal appearance, amodal depth maps, and occlusion statistics (rate and order).

Introducing height offset γ and scaling factor α , the placement plane center and size are

$$c_p = \left(\frac{\min_{c,x} + \max_{c,x}}{2}, \frac{\min_{c,y} + \max_{c,y}}{2}, \min_{c,z} + \gamma \right) \quad (3)$$

$$w_p = \max_{c,x} - \min_{c,x}, \quad l_p = (\max_{c,y} - \min_{c,y}) \cdot \alpha \quad (4)$$

The rectangular region $\mathcal{R}$ is refined by intersecting with the convex hull $\mathcal{H}$ of the xy -projection:

$$Q_c = \{(v_x, v_y) \mid v \in V_c\} \quad (5)$$

$$\mathcal{H} = \text{ConvexHull}(Q_c), \mathcal{P}_{final} = \mathcal{R} \cap \mathcal{H} \quad (6)$$

2.3. Viewpoint Sampling

As shown in Figure 2 (b), from the center of the carrier C , the bounding sphere center and radius are defined as

$$c_s = \frac{1}{2} (\min_{c,x} + \max_{c,x}, \min_{c,y} + \max_{c,y}, \min_{c,z} + \max_{c,z}) \quad (7)$$

$$R_s = \max_{v \in V_c} \|v - c_s\| \quad (8)$$

With offsets β and δ , the hemisphere center and radius are

$$o = c_s + (0, 0, \beta), \quad r = R_s + \delta \quad (9)$$

To prevent uneven sampling, we adopt stratified sampling with elevation θ and azimuth ϕ , introducing $\xi = \sin \theta$. The parameter space is divided into $G \times G$ cells:

$$G = \lceil \sqrt{N} \rceil, \quad \Delta\xi = \frac{\xi_{\max} - \xi_{\min}}{G}, \quad \Delta\phi = \frac{2\pi}{G} \quad (10)$$

Each cell (i, j) yields

$$\theta_{ij} = \arcsin(\xi_{\min} + (i + \eta_i)\Delta\xi), \quad \phi_{ij} = (j + \eta_j)\Delta\phi \quad (11)$$

and the Cartesian coordinates of the viewpoint are

$$x_{ij} = o_x + r \cos \theta_{ij} \cos \phi_{ij} \quad (12)$$

$$y_{ij} = o_y + r \cos \theta_{ij} \sin \phi_{ij} \quad (13)$$

$$z_{ij} = o_z + r \sin \theta_{ij} \quad (14)$$

2.4. 3D Models Placement

MaviGen supports two placement strategies:

(1) Physics-based stacking: Objects are initialized above the placement plane and simulated under gravity, collision, and damping until stable, producing realistic stacks;

(2) Geometry-constrained arrangement: Each object O_i is defined by its 3D vertex set $\mathcal{V}_{O_i}$. We represent its footprint as the convex hull H_{O_i} of the vertices projected onto the xy -plane. Given the geometric centroid c_{O_i} of H_{O_i} , the footprint and the circumscribed radius r_{O_i} (Li et al., 2023) are defined as:

$$H_{O_i} = \text{ConvHull}(\{(v_x, v_y) \mid v \in \mathcal{V}_{O_i}\}) \quad (15)$$

$$r_{O_i} = \max_{(x,y) \in H_{O_i}} \sqrt{(x - c_{O_i,x})^2 + (y - c_{O_i,y})^2} \quad (16)$$

Objects are sorted by r_{O_i} and initialized using a **spiral search** strategy on a polar grid centered at c_p , ensuring $H_{O_i} \subset \mathcal{P}_{final}$. Subsequently, each object is iteratively moved toward c_p for compaction. The movement direction d_{O_i} and the maximum feasible step size $s_{O_i}^*$ (determined via **binary search**) are defined as:

$$d_{O_i} = \frac{c_p - c_{O_i}}{\|c_p - c_{O_i}\|} \quad (17)$$

$$s_{O_i}^* = \max \left\{ s \mid (H_{O_i} + s d_{O_i}) \subset \mathcal{P}_{final}, \right.$$

$$(H_{O_i} + sd_{O_i}) \cap \bigcup_{k \neq i} H_{O_k} = \emptyset \quad (18)$$

This optimization yields compact, non-overlapping configurations, a state we refer to as “close contact without overlap”.

3. EUREKA Model

Figure 3 illustrates EUREKA, a multi-task amodal instance segmentation framework built on a DINO series-pretrained ViT encoder (Oquab et al., 2023). We follow prior work for tokenization and encoder-only refinement with mask-guided self-attention and mask annealing (Keressies et al., 2025). Our main contribution is a *unified query mechanism* with *two task-specific mask heads* (Amodal and Visible) for joint modeling of complete and visible object shapes.

Unified Query Design. Diverging from designs that allocate a set of dedicated queries to specific semantic components (e.g., assigning separate queries for amodal and visible tasks), EUREKA adopts a *Unified Query* mechanism. We initialize a single set of learnable queries $\mathbf{Q} \in \mathbb{R}^{N_q \times d}$. In contrast to decoupled methods that require explicit task-type embeddings to distinguish attributes, our unified queries serve as holistic representations. Through encoder-only interaction, each query $\mathbf{q}_i$ learns to encode both visible appearance cues and amodal structural context within a shared latent space. This design significantly reduces query parameters compared to multi-query baselines and enforces a shared feature representation for both visible and amodal predictions.

Two mask heads with a shared class head. After encoder-only refinement, we obtain refined query tokens $\mathbf{H}^Q \in \mathbb{R}^{B \times N_q \times d}$. We share a classification head across tasks since category prediction is visibility-invariant:

$$\hat{\mathbf{P}} = \text{Cls}(\mathbf{H}^Q), \quad (19)$$

where $\hat{\mathbf{P}} \in \mathbb{R}^{B \times N_q \times (K+1)}$ denotes classification logits (including the “no-object” class). For mask prediction, we employ two independent heads:

$$\mathbf{E}_{\text{mask}}^A = g_A(\mathbf{H}^Q), \quad \mathbf{E}_{\text{mask}}^V = g_V(\mathbf{H}^Q), \quad (20)$$

and obtain the corresponding mask logits by projecting onto the upsampled patch feature map $\mathbf{F}_{\text{up}}$:

$$\hat{\mathbf{M}}^A = \langle \mathbf{E}_{\text{mask}}^A, \mathbf{F}_{\text{up}} \rangle, \quad \hat{\mathbf{M}}^V = \langle \mathbf{E}_{\text{mask}}^V, \mathbf{F}_{\text{up}} \rangle, \quad (21)$$

where $\hat{\mathbf{M}}^A, \hat{\mathbf{M}}^V \in \mathbb{R}^{B \times N_q \times H \times W}$. The Visible Mask Head g_V specializes in extracting reliable visible evidence, while the Amodal Mask Head g_A focuses on recovering complete object extents under occlusion, yielding complementary geometric cues under a shared encoder representation.

Conditional masked self-attention. Following Keressies et al., we predict intermediate masks within the final M encoder layers to constrain query-to-patch interactions. Distinct from the standard single-mask design, EUREKA introduces a conditional mechanism where the attention mask source is selected from *Visible*, *Amodal*, or *Union* (Figure 3 (e)–(g)). We define the binary attention masks as

$$\mathbf{A}_V = \mathbb{I}(\hat{\mathbf{M}}^V > 0), \quad \mathbf{A}_A = \mathbb{I}(\hat{\mathbf{M}}^A > 0), \quad (22)$$

and compute the *Union* attention mask by pixel-wise logical or:

$$\mathbf{A}_{\text{union}} = \mathbf{A}_A \vee \mathbf{A}_V, \quad (23)$$

where $\mathbb{I}(\cdot)$ denotes the binary indicator function. This flexibility allows the model to adapt its receptive field during refinement, enabling queries to capture both direct visible evidence and holistic amodal context.

Paired amodal and visible supervision. For each image I , the dataset provides paired instance-level supervision

$$\mathcal{Y} = \{(y_j, \mathbf{Y}_j^A, \mathbf{Y}_j^V)\}_{j=1}^n, \quad (24)$$

where $y_j \in \{1, \dots, K\}$ denotes the class label and $\mathbf{Y}_j^A, \mathbf{Y}_j^V \in \{0, 1\}^{H \times W}$ are the amodal and visible masks of the same instance.

Multi-task objective. We adopt the mask-classification training paradigm with bipartite matching. To align with the unified query design, EUREKA utilizes *joint Hungarian matching* (Kuhn, 1955). For query i and ground-truth instance j , the joint matching cost is

$$\begin{aligned} \mathcal{C}_{\text{match}}(i, j) &= \lambda_{\text{cls}} \mathcal{C}_{\text{cls}}(\hat{\mathbf{P}}_i, y_j) \\ &+ \sum_{\tau \in \{A, V\}} \left(\lambda_{\text{mask}}^{\tau} \mathcal{C}_{\text{bce}}(\hat{\mathbf{M}}_i^{\tau}, \mathbf{Y}_j^{\tau}) + \lambda_{\text{dice}}^{\tau} \mathcal{C}_{\text{dice}}(\hat{\mathbf{M}}_i^{\tau}, \mathbf{Y}_j^{\tau}) \right), \end{aligned} \quad (25)$$

where $\hat{\mathbf{P}}_i$ and $\hat{\mathbf{M}}_i^{\tau}$ denote the classification logits and mask logits predicted by query i . We then solve for the optimal assignment between the n ground-truth instances and N_q queries:

$$\hat{\pi} = \arg \min_{\pi \in \mathfrak{S}_{N_q}} \sum_{j=1}^n \mathcal{C}_{\text{match}}(\pi(j), j), \quad (26)$$

where $\pi(j) \in \{1, \dots, N_q\}$ denotes the index of the query assigned to ground-truth instance j .

Based on this unified assignment $\hat{\pi}$, we optimize the total multi-task loss

$$\mathcal{L} = \lambda_{\text{cls}} \mathcal{L}_{\text{ce}} + \sum_{\tau \in \{A, V\}} (\lambda_{\text{mask}}^{\tau} \mathcal{L}_{\text{bce}}^{\tau} + \lambda_{\text{dice}}^{\tau} \mathcal{L}_{\text{dice}}^{\tau}), \quad (27)$$

where each loss term is computed over the matched pairs $(\pi(j), j)$. This objective enforces that the same query consistently models both the visible and amodal extents of a target instance.

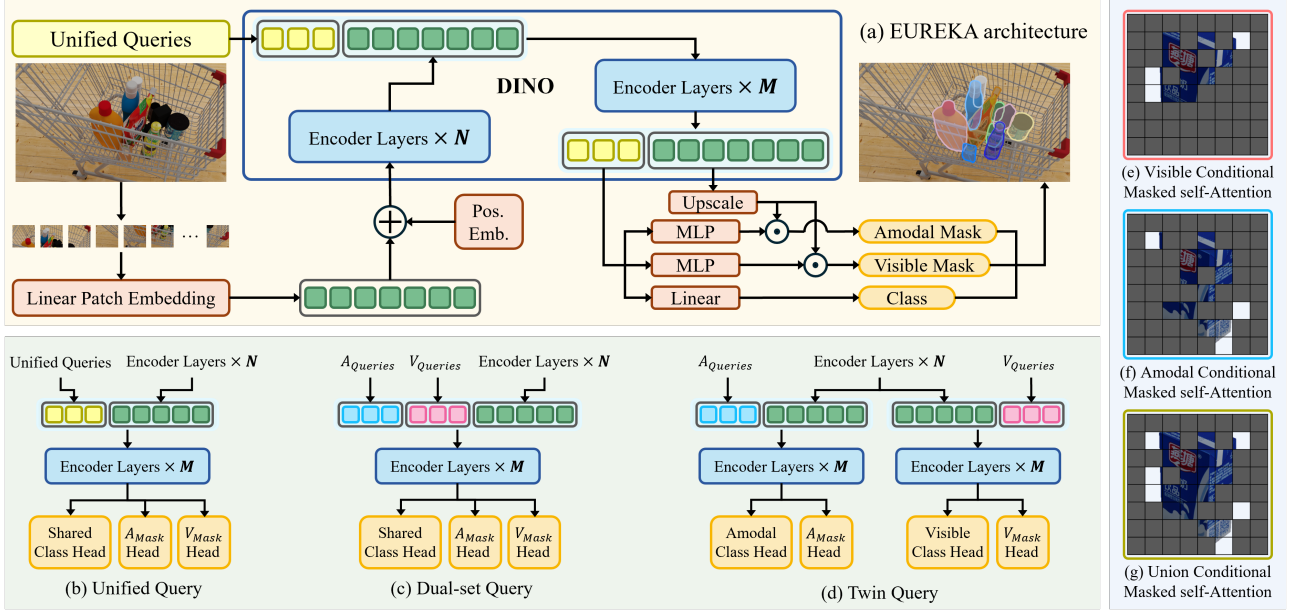


Figure 3. EUREKA architecture. The backbone is a streamlined encoder-only ViT. A unified query mechanism is introduced in the last M encoder layers, jointly encoding appearance cues and structural context in a shared latent space to avoid feature fragmentation induced by separate queries. The segmentation heads, including a shared classification head, an amodal mask head, and a visible mask head, are shared to enable multi-task outputs. For mask-guided attention, three conditioning choices are compared: visible masks, modality-agnostic masks, and union masks. In particular, the union mask exposes both the visible region and the complete silhouette, facilitating occlusion reasoning and learning the logic of amodal completion. The input image is divided into patches via linear patch embedding, then combined with learnable positional embeddings (Pos. Emb.) to preserve spatial information.

4. Experiments

4.1. Experimental Settings

Experiments are conducted on 3D-IRAIS with a 6:2:2 train, validation, and test split, and cross-tested on Real-IRAIS, which contains 520 manually annotated images consistent with 3D-IRAIS in product categories and scene environments to quantify the sim-to-real gap. We instantiate the ViT backbone using DINOv2 ViT-L/14 and DINOv3 ViT-L/16 pre-trained weights, respectively. All models are trained using the AdamW optimizer with an initial learning rate of 1×10^{-4} . To effectively adapt the pre-trained ViT encoder, we apply a layer-wise learning rate decay of 0.8, along with a polynomial learning rate scheduler and polynomial mask annealing, both with a decay power of 0.9. For the multi-task objective and bipartite matching cost, we set the class coefficient λ_{cls} to 2.0 and the no-object coefficient λ_{no_object} to 0.1; for both amodal and visible mask predictions, we use a BCE weight λ_{mask}^A and λ_{mask}^V of 5.0, and a Dice loss weight λ_{dice}^A and λ_{dice}^V of 5.0. The input resolution is fixed at 640×640 , utilizing a Shortest-Edge Scale Jittering (SSJ) strategy that resizes images based on the shortest edge while preserving the aspect ratio followed by padding. Training is conducted for 89k iterations on 4 NVIDIA A800 GPUs with a batch size of 2. We report occlusion-aware AP and AR, as well as per-category segmentation performance.

Table 1. Comparison of Existing Amodal Datasets.

Dataset	Domain	Images	Instances	Categories	Mask	DM	OO	CP
COCOA	Common	5,073	46,314	80	✓	✗	✗	✗
KINS	Vehicle	14,991	190,626	8	✓	✗	✗	✗
D2SA	Shopping	5,600	28,720	19	✓	✗	✗	✗
ORFD	Retail	1,000	10,000	200	✗	✗	✗	✗
MUVA	Indoor	26,406	198,573	20	✓	✓	✗	✗
DYCE	Indoor	5,500	85,975	79	✓	✗	✗	✗
OLMD	Indoor	13,000	—	40	✓	✗	✓	✗
3D-IRAIS	Retail	6,016 $\rightarrow \infty$	61,909 $\rightarrow \infty$	300+	✓	✓	✓	✓
Real-IRAIS	Retail	235	2605	42	✓	✗	✓	✗

✓ denotes presence; ✗ denotes absence. DM: Depth Maps; OO: Occlusion Order; CP: Camera Parameters. ∞ : Indicates that with MaviGen, the dataset can be scaled indefinitely beyond the current 3D-IRAIS release.

4.2. Data Generation and Dataset Statistics

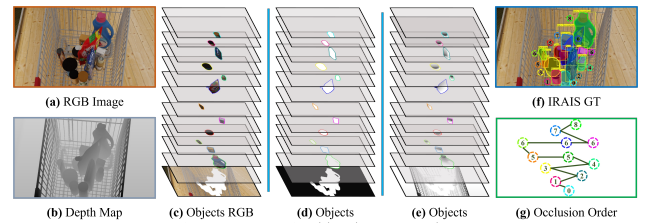


Figure 4. Amodal multi-modal analysis for occluded scenes.

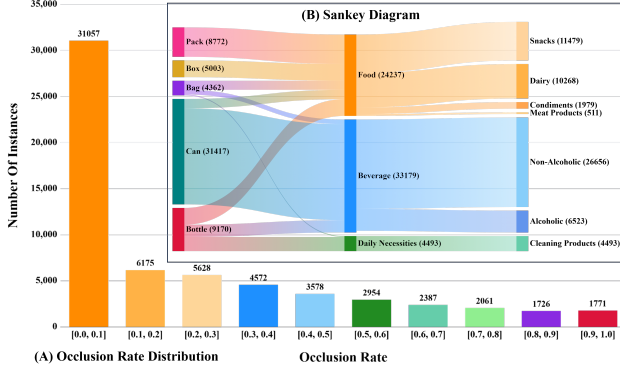


Figure 5. Statistical overview of 3D-IRAIIS.

For each viewpoint, MaviGen generates multimodal amodal annotations using the Blender physics and rendering engines by integrating RGB rendering, depth estimation, object-level appearance modeling, and occlusion reasoning. As shown in Figure 4, the system renders the full scene into an RGB image and a depth map, and further renders each object individually to obtain per-object RGB appearance, amodal masks capturing complete silhouettes, and per-object depth. These outputs are consolidated into IRAIS annotations in a unified format, which aggregates visible masks, amodal masks, and depth. Finally, an occlusion-order graph (Figure 4 (g)) is derived by analyzing depth relations and mask overlaps, providing explicit occlusion-order supervision for amodal segmentation. All annotations are stored in a standardized directory structure, with a centralized label file recording camera parameters, image resolution, object-level 2D/3D statistics, and file paths, enabling efficient storage, training, and evaluation.

Table 1 compares 3D-IRAIIS with existing amodal segmentation datasets. Unlike prior datasets providing only amodal masks, 3D-IRAIIS includes depth maps, occlusion order, and camera parameters, enabling amodal segmentation with explicit geometric priors. As shown in Figure 5, the dataset exhibits a long-tailed occlusion distribution with balanced category coverage. 3D-IRAIIS offers three key advantages: (1) scalability through the fully automated MaviGen pipeline, enabling unlimited data generation; (2) multi-modal richness for exploring various geometric prior combinations; and (3) unified benchmarking with consistent annotation quality for fair comparison across methods.

4.3. Main Results and Analysis

Benchmark Analysis. In Table 2, our EUREKA achieves the best multi-task performance on 3D-IRAIIS, improving both amodal and visible segmentation under the same model. With DINO backbones, EUREKA attains 89.10%/89.28% (amodal/visible AP) on DINOv2 and further reaches 89.75% amodal AP with 90.57% visible AP on DINOv3 (Siméoni

et al., 2025), surpassing all two-task baselines and establishing a strong in-domain upper bound. More importantly, Table 3 demonstrates that these gains transfer reliably to real-world retail images: when trained only on 3D-IRAIIS and evaluated on Real-IRAIIS, EUREKA delivers 37.10% amodal AP and 41.95% visible AP (DINOv2), dramatically outperforming prior methods (typically 10%–23% AP) by a large margin. This consistent improvement across all metrics indicates a substantially reduced sim-to-real generalization gap, validating that our paired amodal/visible learning and encoder-only design enable effective transfer from virtual scenes to real retail environments.

Table 4 and Table 5 compare EUREKA with representative amodal instance segmentation baselines on two public datasets, including D2SA and COCOA-clis (real-world open scenes). On D2SA, EUREKA establishes a new state of the art, achieving 84.84% amodal AP and 89.78% visible AP, exceeding the strongest prior result (AISFormer+SAMEO, 74.1% amodal AP and 79.8% visible AP) by 10.7 and 10.0 AP, respectively. On COCOA-clis, despite the domain shift to diverse real-world scenes, EUREKA remains consistently superior, improving amodal AP to 39.50% and visible AP to 40.59%, while also delivering substantially higher recall (amodal AR 52.63%, visible AR 53.62%).

Ablation Study. Table 6 compares three query mechanisms for end-to-end joint amodal/visible segmentation. The *Unified Query* paradigm delivers the best overall accuracy while being more efficient: it achieves 89.10% amodal AP and 89.28% visible AP with 690 GFLOPs and 315M parameters, outperforming Dual-set Query and Twin Query (which is notably heavier at 846 GFLOPs / 420M), indicating that sharing instance queries provides stronger coupling between amodal completion and visible evidence.

Table 7 summarizes the model design options for EUREKA across tasks, enabling flexible adaptation rather than a one-size-fits-all configuration. We compare three mask-attention training schemes, namely *visible-only*, *amodal-only*, and *union*, under the same Unified Query setting across representative backbones and evaluation sets, and observe a clear dataset- and task-dependent preference. In particular, *Union* attention is generally more effective for heavily occluded retail scenes with concentrated occlusions and stacked instances (e.g., 3D-IRAIIS), as it exposes the attention module to both visible evidence and completed regions and thus better matches the underlying occlusion structure. Conversely, for datasets with more diverse occlusion patterns and open-world semantics (e.g., COCOA-clis), *Amodal-focused* attention can be the more suitable choice, suggesting that emphasizing amodal reasoning is beneficial when the visible evidence is less structured and the completion cues are more heterogeneous. Overall, Table 7 highlights that EUREKA supports multiple task-specific adaptation recipes:

Amodal Instance Segmentation with IRAIS Dataset for Sim-to-Real Transfer

Table 2. Comprehensive and detailed comparison of amodal and visible segmentation performance on the 3D-IRAIS dataset across different backbone settings (ResNet-50/101 and DINOv2/v3). The results indicate that our 3D modeling achieves performance comparable to D2SA. FPS and Forward (ms) report inference throughput (frames per second) and per-image forward-pass latency (milliseconds), respectively, measured on a single NVIDIA RTX A5000 GPU, and are reported for the amodal modality branch.

Model	Backbone	Amodal (%)				Visible (%)				FPS	Forward (ms)
		AP	AP ₅₀	AP ₇₅	AR ₁₀₀	AP	AP ₅₀	AP ₇₅	AR ₁₀₀		
APNet	ResNet-50	76.85	94.04	86.50	81.74	77.19	93.76	85.98	85.42	30	33.84
VRSP-Net		77.62	92.16	84.68	85.81	76.72	90.92	82.80	85.42	9	112.83
ORCNN		81.07	93.99	89.16	85.61	81.36	93.54	88.06	86.04	28	36.20
Mask R-CNN		81.79	94.03	89.91	86.36	—	—	—	—	28	35.10
AISFormer		82.15	94.62	90.16	86.42	83.47	93.95	89.87	86.68	30	33.75
APNet	ResNet-101	77.81	93.44	86.10	82.72	44.90	72.74	42.66	60.04	24	42.22
VRSP-Net		80.02	93.39	86.50	85.88	78.63	91.14	84.27	85.44	8	128.38
ORCNN		81.88	94.01	89.72	86.38	81.98	93.62	88.85	86.70	22	45.54
Mask R-CNN		81.86	94.00	89.68	86.69	—	—	—	—	22	44.63
AISFormer		82.67	94.65	90.13	86.44	83.17	93.99	89.76	87.08	24	41.34
EUREKA (ours)	DINOv2	89.10	98.13	95.43	90.50	89.28	97.97	94.22	90.29	21	47.97
	DINOv3	89.75	98.24	95.34	91.20	90.57	97.98	95.60	91.75	16	61.45

“—” indicates missing visible metrics since these models are single-task; the same applies to other tables.

Table 3. **Sim-to-real:** train on **3D-IRAIS**, test on **Real-IRAIS**. Real-IRAIS reflects real retail environments with diverse lighting, reflections, and cluttered backgrounds. It also contains frequent occlusions and crowded layouts, posing practical challenges for amodal and visible segmentation. Real-IRAIS exhibits real-world environmental factors similar to those observed in COCOA-clis.

Model	Backbone	Amodal (%)				Visible (%)			
		AP	AP ₅₀	AP ₇₅	AR ₁₀₀	AP	AP ₅₀	AP ₇₅	AR ₁₀₀
APNet	ResNet-50	29.19	46.36	31.83	49.79	23.53	42.54	22.74	44.10
VRSP-Net		16.09	26.72	16.88	35.21	16.48	24.50	18.53	34.93
ORCNN		18.67	32.33	19.41	35.86	21.17	31.69	23.37	37.92
Mask R-CNN		18.25	30.93	19.59	34.04	—	—	—	—
AISFormer		17.84	29.43	18.86	35.31	20.53	29.12	23.39	37.88
APNet	ResNet-101	18.66	33.28	18.99	36.75	14.84	29.88	13.21	32.65
VRSP-Net		16.95	27.12	18.68	37.62	18.51	26.45	20.19	37.37
ORCNN		19.76	32.46	21.06	37.17	23.02	33.14	24.99	40.27
Mask R-CNN		18.48	30.92	19.54	34.65	—	—	—	—
AISFormer		18.82	30.53	19.79	36.41	21.69	31.03	23.75	39.00
EUREKA (ours)	DINOv2	37.10	50.71	40.34	55.99	41.95	51.88	45.35	60.83
	DINOv3	33.00	46.59	34.56	56.03	38.46	48.55	39.31	60.49

Table 4. Amodal and visible segmentation results on **D2SA**. Due to the high annotation cost of Real-IRAIS, we leverage the larger-scale D2SA to further validate EUREKA’s generalization capability.

Method	Year	Shape Prior	Amodal(%)				Visible(%)			
			AP	AP ₅₀	AP ₇₅	AR	AP	AP ₅₀	AP ₇₅	AR
Mask R-CNN	ICCV17	✗	63.57	83.85	68.02	65.18	68.98	—	—	70.11
ASN [†]	CVPR19	✗	63.94	84.35	69.57	65.20	—	—	—	—
ORCNN	WACV19	✗	64.22	83.55	69.12	65.25	69.67	—	—	70.46
Mask2Former	CVPR22	✗	65.32	82.10	70.55	66.45	—	—	—	—
BCNet [†]	CVPR21	✗	65.97	84.23	72.74	66.90	—	—	—	—
AISFormer	BMVC22	✗	67.22	84.05	72.87	68.13	71.60	—	—	71.59
RTMDet*+SAMEO [†]	CVPR2025	✓	68.4	84.5	71.0	77.7	72.7	85.8	77.5	84.2
VRSP-Net	AAAI21	✓	70.27	85.11	75.81	69.17	72.28	—	—	71.85
AISFormer+SAMEO [†]	CVPR2025	✗	74.1	90.2	78.3	79.9	79.8	92.7	84.9	84.2
EUREKA-DINOv2(Ours)	ICML2026	✗	82.35	91.77	87.10	88.10	86.94	92.21	90.41	92.36
EUREKA-DINOv3(Ours)	ICML2026	✗	84.84	93.5	88.53	89.99	89.78	93.81	92.12	94.61

[†] Results reproduced or cited from original papers. ✓ indicates methods utilizing shape prior; ✗ indicates methods not utilizing shape prior.

Table 5. Amodal and visible segmentation results on **COCOA-clS** (a real-world scenario dataset).

Method	Year	Shape Prior	Amodal(%)				Visible(%)			
			AP	AP ₅₀	AP ₇₅	AR	AP	AP ₅₀	AP ₇₅	AR
Mask R-CNN	ICCV17	✗	33.67	56.50	35.78	34.18	30.10	—	—	31.52
ORCNN	WACV19	✗	28.03	53.68	25.36	29.83	30.80	—	—	32.23
ASN [†]	CVPR19	✗	35.33	58.82	37.10	35.50	—	—	—	—
BCNet [†]	CVPR21	✗	35.14	58.84	36.65	35.80	—	—	—	—
VRSP-Net	AAAI21	✓	35.41	56.03	38.67	37.11	34.58	—	—	36.42
AISFormer	BMVC22	✗	35.77	57.95	38.23	36.71	34.00	—	—	36.44
Mask2Former	CVPR22	✗	34.20	56.15	36.80	35.20	—	—	—	—
Shapeformer [†]	IJCNN2024	✓	35.83	58.82	38.85	37.13	35.01	—	—	36.42
EUREKA-DINOv2(Ours)	ICML2026	✗	37.82	50.64	41.65	54.16	39.30	51.02	43.62	55.88
EUREKA-DINOv3(Ours)	ICML2026	✗	39.50	49.55	43.77	52.63	40.59	49.62	45.52	53.62

Table 6. Performance Comparison of Different Query Mechanisms on the 3D-IRAIS Dataset for Amodal Instance Segmentation.

Paradigms	Amodal(%)				Visible(%)				GFLOPs	Params
	AP	AP ₅₀	AP ₇₅	AR	AP	AP ₅₀	AP ₇₅	AR		
Unified Query	89.10	98.13	95.43	90.50	89.28	97.97	94.22	90.29	690	315M
Dual-set Query	88.88	98.17	94.90	90.21	88.98	97.58	93.88	89.87	701	318M
Twin Query	88.49	98.21	94.69	89.86	88.93	97.62	93.91	89.79	846	420M

Table 7. EUREKA performance under different conditional masked self-attention, backbones, and test datasets, providing guidance for flexible multi-task training and backbone selection.

Mask Conditioning			Backbone	Dataset	Amodal(%)				Visible(%)			
Amodal	Visible	Union			AP	AP ₅₀	AP ₇₅	AR ₁₀₀	AP	AP ₅₀	AP ₇₅	AR ₁₀₀
✓			DINOv2	3D-IRAIS	88.90	98.31	95.16	90.31	88.86	97.98	93.57	89.92
	✓		DINOv2		89.10	98.13	95.43	90.50	89.28	97.97	94.22	90.29
		✓	DINOv2		88.70	97.89	94.85	90.17	88.93	97.65	94.02	89.99
✓			DINOv3		89.49	98.01	94.98	91.04	90.44	97.65	95.30	91.61
	✓		DINOv3		89.19	97.78	94.90	90.66	90.33	97.65	95.12	91.36
		✓	DINOv3		89.75	98.24	95.34	91.20	90.57	97.98	95.60	91.75
✓			DINOv2	COCOA-ClS	37.56	50.11	41.70	53.95	39.33	50.47	44.34	55.78
	✓		DINOv2		37.82	50.64	41.65	54.16	39.30	51.02	43.62	55.88
		✓	DINOv2		37.73	49.79	42.21	52.53	39.53	50.53	44.49	54.52
✓			DINOv3		39.50	49.55	43.77	52.63	40.59	49.62	45.52	53.62
	✓		DINOv3		38.55	48.96	42.61	53.34	39.48	48.99	44.26	54.80
		✓	DINOv3		38.65	49.13	42.60	52.77	39.60	49.35	44.00	53.91

Visible/Amodal/Union denote the Visible, Amodal, and Union Masked self-Attention, respectively. ✓ means the component is used.

practitioners can select the attention scheme and backbone pair that best matches the target data distribution and task requirements, and verify the choice via the same ablation protocol on the target benchmark. Our model generalizes well across datasets with distinct image characteristics and occlusion patterns. As shown in Figure 1, it produces coherent amodal instance masks in crowded retail settings (3D-IRAIS, Real-IRAIS) and maintains stable performance on diverse everyday scenes (D2SA, COCOA-clS), indicating strong robustness to clutter and heavy occlusions.

Evaluation stability under different real-data scales. All models are trained on the same 3D-IRAIS training set.

We assess cross-domain generalization on two real-image evaluation sets with different scales: the original few-shot Real-IRAIS with 235 images and 2,605 instances (Table 2), and the expanded Real-IRAIS-Aug with 1,175 images and 13,025 instances (Table 8). Real-IRAIS-Aug is obtained by applying label-preserving geometric augmentations to Real-IRAIS, including random horizontal flipping and symmetric transformations (e.g., rotations), which increases the evaluation size without introducing new annotations. This protocol enables a controlled comparison of whether the performance trends observed on the 235-image Real-IRAIS split remain consistent when the evaluation scale is enlarged.

Table 8. Small-sample stability evaluation. We augment the few-shot Real-IRAIS set to construct Real-IRAIS-Aug. Using the model weights trained on 3D-IRAIS, we perform cross-domain testing on Real-IRAIS-Aug to quantify performance variance and assess the robustness and stability of each model under limited real-world data. FPS and Forward (ms) report inference throughput (frames per second) and per-image forward-pass latency (milliseconds), respectively, measured on the amodal modality branch on one NVIDIA A800 GPU.

Model	Backbone	Amodal				Visible				FPS	Forward (ms)
		AP	AP ₅₀	AP ₇₅	AR ₁₀₀	AP	AP ₅₀	AP ₇₅	AR ₁₀₀		
APNet	ResNet-50	26.60	42.97	28.58	47.33	22.47	40.18	22.02	42.93	49	20.40
VRSP-Net		14.15	24.02	15.00	34.35	13.45	21.32	14.39	31.79	7	146.26
ORCNN		16.17	28.49	16.41	32.81	18.06	27.87	19.30	34.63	43	23.17
Mask R-CNN		15.87	27.80	16.28	31.62	—	—	—	—	43	23.05
AISFormer		15.52	26.44	15.78	32.84	17.38	25.94	18.92	34.33	36	27.84
APNet	ResNet-101	16.48	29.45	16.52	34.27	13.92	27.40	12.77	31.31	40	25.28
VRSP-Net		15.04	24.68	16.09	35.36	16.05	23.98	17.10	34.63	6	158.13
ORCNN		17.30	29.08	17.93	34.63	19.80	29.14	21.41	37.24	33	30.18
Mask R-CNN		15.91	27.57	16.31	31.78	—	—	—	—	34	29.20
AISFormer		17.16	28.22	18.02	34.01	19.40	28.35	21.01	36.17	32	31.22
EUREKA	DINOv2	34.87	47.98	38.09	54.28	39.39	49.51	42.10	58.58	39	25.61
EUREKA	DINOv3	32.00	44.96	33.84	54.83	37.84	47.62	38.65	59.47	29	34.92

Quantifying fluctuations. As shown in Table 8. To make the comparison explicit, we treat the scores on Real-IRAIS and Real-IRAIS-Aug as two observations and report their dispersion using the two-point population standard deviation σ and sample standard deviation s . For EUREKA with DINOv2, the amodal AP changes from 37.10 to 34.87, yielding $\sigma \approx 1.12$ and $s \approx 1.58$; the visible AP changes from 41.95 to 39.39, yielding $\sigma \approx 1.28$ and $s \approx 1.81$. For EUREKA with DINOv3, the amodal AP changes from 33.00 to 32.00, corresponding to $\sigma = 0.50$ and $s \approx 0.71$, while the visible AP changes from 38.46 to 37.84, giving $\sigma \approx 0.31$ and $s \approx 0.44$.

Although this two-point estimate does not substitute for multi-seed variance analysis, it serves as a lightweight stability check: the performance differences between Real-IRAIS and Real-IRAIS-Aug remain small, indicating consistent behavior across the two evaluation scales. Moreover, the larger evaluation set reduces evaluation noise and provides a more reliable estimate of cross-domain performance, while also exposing the expected synthetic-to-real domain gap when transferring from 3D-IRAIS training data to real-world images. Combined with EUREKA’s strong results on public benchmarks (e.g., COCOA-cls and D2SA), the Real-IRAIS/Real-IRAIS-Aug results show consistent trends across evaluation settings, supporting the conclusion that EUREKA achieves competitive accuracy and stable amodal and visible segmentation performance in low-data real-world scenarios.

5. Conclusion

We propose IRAIS, a sim-to-real amodal instance segmentation benchmark for retail scenarios comprising 3D-IRAIS

and Real-IRAIS. 3D-IRAIS is generated using MaviGen and provides scalable multi-view synthetic data with depth information, occlusion ordering, and camera parameters, offering a multimodal data foundation for researchers to implement and compare methods under the same task setting. Real-IRAIS is a human-annotated high-quality few-shot test set collected from real scenes, enabling sim-to-real transfer evaluation under a unified annotation protocol. We introduce EUREKA, a lightweight model that performs occlusion reasoning using only a single ViT encoder, a unified query mechanism, and a dual-mask head. Experiments show strong in-domain performance on 3D-IRAIS, excellent transferability to Real-IRAIS, and state-of-the-art results on D2SA and COCOA-cls.

The dataset-model co-design strategy demonstrates that geometry-aware supervision from synthetic data can effectively bridge the sim-to-real gap, reducing reliance on costly real-world annotations. We hope our benchmark and model contribute to amodal perception tasks and inspire future research on scalable data generation for occlusion reasoning.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Ao, J., Ke, Q., and Ehinger, K. A. Amodal intra-class instance segmentation: Synthetic datasets and benchmark. In *Proceedings of the IEEE/CVF Winter Conference on*

- Applications of Computer Vision*, pp. 281–290, 2024.
- Cai, T. T. and Ma, R. Theoretical foundations of t-sne for visualizing high-dimensional clustered data. *Journal of Machine Learning Research*, 23(301):1–54, 2022.
- Chen, K., Ramanan, D., and Khurana, T. Using diffusion priors for video amodal segmentation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 22890–22900, 2025.
- Chen, W., Liu, Y., Wang, W., Bakker, E. M., Georgiou, T., Fieguth, P., Liu, L., and Lew, M. S. Deep learning for instance retrieval: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(6):7270–7292, 2022.
- Follmann, P., Bottger, T., Hartinger, P., König, R., and Ulrich, M. Mvtec d2s: densely segmented supermarket dataset. In *Proceedings of the European conference on computer vision (ECCV)*, pp. 569–585, 2018.
- Follmann, P., König, R., Härtinger, P., Klostermann, M., and Böttger, T. Learning to see the invisible: End-to-end trainable amodal instance segmentation. In *2019 IEEE Winter Conference on Applications of Computer Vision (WACV)*, pp. 1328–1336. IEEE, 2019.
- Geng, W., Han, F., Lin, J., Zhu, L., Bai, J., Wang, S., He, L., Xiao, Q., and Lai, Z. Fine-grained grocery product recognition by one-shot learning. In *Proceedings of the 26th ACM international conference on Multimedia*, pp. 1706–1714, 2018.
- He, K., Gkioxari, G., Dollár, P., and Girshick, R. Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision*, pp. 2961–2969, 2017.
- Ke, L., Tai, Y.-W., and Tang, C.-K. Deep occlusion-aware instance segmentation with overlapping bilayers. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 4019–4028, 2021.
- Kerssies, T., Cavagnero, N., Hermans, A., Norouzi, N., Averta, G., Leibe, B., Dubbelman, G., and de Geus, D. Your vit is secretly an image segmentation model. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 25303–25313, 2025.
- Kuhn, H. W. The hungarian method for the assignment problem. *Naval research logistics quarterly*, 2(1-2):83–97, 1955.
- Li, Z., Ye, W., Terven, J., Bennett, Z., Zheng, Y., Jiang, T., and Huang, T. Muva: A new large-scale benchmark for multi-view amodal instance segmentation in the shopping scenario. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 23504–23513, 2023.
- Mohan, R. and Valada, A. Amodal panoptic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 21023–21032, 2022.
- Mondelli, M. and Venkataramanan, R. Pca initialization for approximate message passing in rotationally invariant models. *Advances in Neural Information Processing Systems*, 34:29616–29629, 2021.
- Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.
- Qi, L., Jiang, L., Liu, S., Shen, X., and Jia, J. Amodal instance segmentation with kins dataset. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3014–3023, 2019.
- Siméoni, O., Vo, H. V., Seitzer, M., Baldassarre, F., Oquab, M., Jose, C., Khalidov, V., Szafraniec, M., Yi, S., Ramamonjisoa, M., et al. Dinov3. *arXiv preprint arXiv:2508.10104*, 2025.
- Tai, W.-E., Shih, Y.-L., Sun, C., Wang, Y.-C. F., and Chen, H.-T. Segment anything, even occluded. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 29385–29394, 2025.
- Tran, M., Vo, K., Yamazaki, K., Fernandes, A., Kidd, M., and Le, N. Aisformer: Amodal instance segmentation with transformer. *arXiv preprint arXiv:2210.06323*, 2022.
- Tran, M., Bounsavy, W., Vo, K., Nguyen, A., Nguyen, T., and Le, N. Shapeformer: Shape prior visible-to-amodal transformer-based amodal instance segmentation. In *2024 international joint conference on neural networks (IJCNN)*, pp. 1–8. IEEE, 2024.
- Xiao, Y., Xu, Y., Zhong, Z., Luo, W., Li, J., and Gao, S. Amodal segmentation based on visible region segmentation and shape prior. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pp. 2995–3003, 2021.
- Zhang, G., Navasardyan, S., Chen, L., Zhao, Y., Wei, Y., Shi, H., et al. Mask matching transformer for few-shot segmentation. *Advances in Neural Information Processing Systems*, 35:823–836, 2022.
- Zhu, Y., Tian, Y., Metaxas, D., and Dollár, P. Semantic amodal segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 1464–1472, 2017.

A. Appendix

A.1. Camera Matrix and Formation

MaviGen supports a collaborative camera matrix consisting of multiple cameras that can be flexibly organized to meet diverse acquisition requirements. Three core layouts are provided: linear array, circular array, and grid array, each optimized for specific retail scenarios. (1) Linear array: Cameras are aligned along a predefined axis with fixed baseline spacing. This layout enables continuous sampling of elongated regions, such as self-checkout counters. (2) Circular array: Cameras are distributed in an arc or closed loop around the target, supporting surround sampling and environmental perception for retail robots. (3) Grid array: A 2D grid-based camera arrangement provides dense coverage over wide areas. This arrangement can be configured to balance sampling resolution against field-of-view coverage and to focus on regions of interest.

A.2. Geometry-Constrained Arrangement Algorithm

To ensure the reproducibility of our placement strategy, we provide the detailed pseudocode for the geometry-constrained arrangement in Algorithm 1. This process corresponds to the methodology described in section 2.4, comprising two distinct stages: Spiral Initialization and Centripetal Compaction.

Algorithm 1 Geometry-Constrained Arrangement

Require: Object set $\mathcal{O} = \{O_1, \dots, O_N\}$, Plane Center c_p , Valid Region $\mathcal{P}_{final}$

Require: Parameters: Radial step $\Delta\rho$, Angular step $\Delta\phi$, Iterations T

Stage 1: Spiral Initialization

Sort $\mathcal{O}$ by descending circumscribed radius r_{O_i}

Initialize placed objects set $\mathcal{S} \leftarrow \emptyset$

for each object $O_i \in \mathcal{O}$ **do**

$\rho \leftarrow 0$

$is_placed \leftarrow \text{false}$

while not is_placed **and** $\rho < \rho_{max}$ **do**

for $\phi = 0$ to 2π step $\Delta\phi$ **do**

 Calculate candidate centroid $c_{temp} = c_p + (\rho \cos \phi, \rho \sin \phi)$

 Construct footprint H_{O_i} at c_{temp} with random yaw ψ

if $H_{O_i} \subset \mathcal{P}_{final}$ **and** $H_{O_i} \cap (\bigcup_{H_k \in \mathcal{S}} H_k) = \emptyset$ **then**

 Update O_i pose to (c_{temp}, ψ)

$\mathcal{S} \leftarrow \mathcal{S} \cup \{H_{O_i}\}$

$is_placed \leftarrow \text{true}$

break

end if

end for

$\rho \leftarrow \rho + \Delta\rho$ {Expand search radius outward}

end while

end for

Stage 2: Centripetal Compaction

for iteration $t = 1$ to T **do**

for each object $O_i \in \mathcal{O}$ **do**

 Calculate centripetal direction $\vec{d}_{O_i} \leftarrow (c_p - c_{O_i}) / \|c_p - c_{O_i}\|$

 {Find max step s^* via binary search satisfying Eq. 18}

$s^* \leftarrow \text{BinarySearch}(O_i, \vec{d}_{O_i}, \mathcal{P}_{final}, \mathcal{S} \setminus \{H_{O_i}\})$

 Update position: $c_{O_i} \leftarrow c_{O_i} + s^* \vec{d}_{O_i}$

 Update footprint H_{O_i} in $\mathcal{S}$

end for

end for

return Final poses of $\mathcal{O}$

A.3. Parameter Configuration for the IRAIS Dataset Generation

To ensure the reproducibility of the dataset generation process described in sections 2.2 and section 2.3, we provide the specific parameter settings used to generate the **IRAIS** dataset. Note that these values were optimized based on the scale and geometry of the specific retail assets and carriers employed in our experiments.

Geometric Configuration. For the placement plane, we empirically selected a height offset of $\gamma = 0.005$ m to prevent mesh interpenetration while ensuring stable contact with the diverse carrier surfaces in our library. A scaling factor of $\alpha = 0.8$ was applied to the carrier’s bounding box dimensions to guarantee that initialized objects remain safely within the container boundaries during the physical settling phase. The placement center and region dimensions are defined as follows (reproduced from Equation 4 and Equation 3 for convenience):

$$c_p = \left(\frac{\min_{c,x} + \max_{c,x}}{2}, \frac{\min_{c,y} + \max_{c,y}}{2}, \min_{c,z} + \gamma \right), \quad w_p = \max_{c,x} - \min_{c,x}, \quad l_p = (\max_{c,y} - \min_{c,y}) \cdot \alpha$$

Sampling Parameters. The IRAIS dataset was generated with $N = 2000$ sampled viewpoints per carrier configuration, a density chosen to balance coverage with computational efficiency and to match the practical capture limit imposed by the resolution configuration of our *Skyworth 3D Otter Lite* scanner. The stratified sampling grid is bounded within an elevation range of $\theta \in [30^\circ, 80^\circ]$ (corresponding to $\xi_{\min} = \sin(30^\circ)$ and $\xi_{\max} = \sin(80^\circ)$) to avoid degenerate views near the poles or occlusion at grazing angles. For readability, we reproduce the viewing angle computation from Equation 11:

$$\theta_{ij} = \arcsin(\xi_{\min} + (i + \eta_i)\Delta\xi), \quad \phi_{ij} = (j + \eta_j)\Delta\phi$$

where the jitter variables $\eta_i, \eta_j \sim \mathcal{U}[0, 1]$ ensure stochastic coverage within each stratified cell.

A.4. Detailed Experimental Settings

To aid reproducibility, we provide specific values for the architecture configurations and loss hyperparameters omitted in the main text.

Network Architecture Parameters. Unless otherwise specified, we follow the default configurations of the DINO-pretrained ViT backbone. In accordance with EoMT (Kerssies et al., 2025), we instantiate $N_q = 200$ learnable unified queries for the amodal instance segmentation task. For the ViT-Large backbone, which comprises a total of 24 transformer encoder layers, we apply the unified query interaction mechanism within the final $M = 4$ layers, while the preceding $N = 20$ layers function as standard encoder blocks.

A.5. Visual Evidence for the Query Mechanism

To qualitatively verify that the decoder *queries* serve as structured instance representations (rather than arbitrary slots), we visualize the manifold of query embeddings and inspect whether it exhibits meaningful organization with respect to both *semantic identity* and *occlusion severity*. The key hypothesis is that a *unified* query representation has sufficient representational capacity to simultaneously encode (i) *what the object is* (semantics) and (ii) *how visible it is* (occlusion state), which is crucial for handling non-trivial occlusion patterns.

Feature extraction and alignment. We run inference on the validation/test split and extract the last-layer decoder query embeddings $q \in \mathbb{R}^D$. Predicted queries are associated with ground-truth instances via Hungarian matching, and we keep only matched *positive queries* to avoid background “no-object” queries affecting the analysis.

Ground-truth occlusion ratio. For each matched instance, we compute a continuous ground-truth occlusion ratio $r_{\text{occ}} \in [0, 1]$ and use it as the color signal:

$$r_{\text{occ}} = 1 - \frac{\text{Area}(\text{Mask}_{\text{visible}})}{\text{Area}(\text{Mask}_{\text{amodal}})} \quad (28)$$

where $\text{Mask}_{\text{visible}}$ denotes the visible-region mask and $\text{Mask}_{\text{amodal}}$ denotes the full (amodal) mask. Larger values indicate heavier occlusion.

Manifold projection. Since Transformer features are often compared by angular similarity, we apply ℓ_2 normalization to suppress magnitude effects:

$$\hat{q} = \frac{q}{\|q\|_2} \quad (29)$$

We then project $\hat{q}$ to 2D using t-SNE (Cai & Ma, 2022) with PCA initialization (Mondelli & Venkataramanan, 2021). To obtain stable neighborhood structures across classes of different sample sizes, we adopt a class-wise perplexity strategy (larger perplexity for large- N classes to emphasize global topology, and smaller perplexity for small- N classes to preserve local structures). We stress that t-SNE axes have no physical meaning; we interpret the visualization through cluster/island formation and the continuity of the occlusion-color distribution.

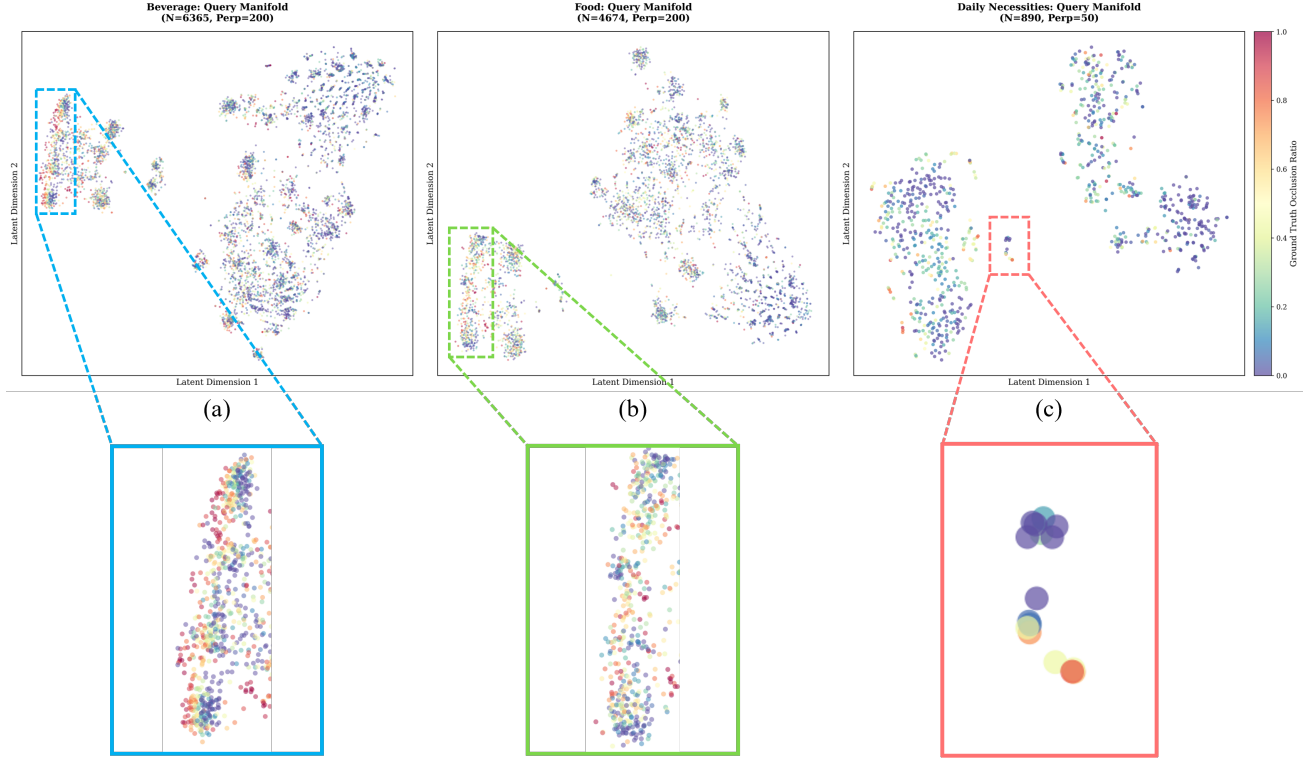


Figure 6. t-SNE visualization of ℓ_2 -normalized decoder query embeddings for different categories. Each dot denotes a Hungarian-matched positive query and is colored by the ground-truth occlusion ratio r_{occ} in Eq. (28). The projection forms multiple compact “islands” (local clusters), suggesting that unified queries capture structured instance semantics. Zoom-in views further reveal non-random and often smooth color variations within islands (e.g., the vertical gradients in a and b), indicating that occlusion severity is encoded as a continuous factor in the learned query representations.

Observations: hierarchical structure in the query space. As shown in Figure 6, the query embeddings exhibit a clear *hierarchical encoding* pattern:

- **Semantic clustering (inter-island separation).** Queries form multiple separated “islands” rather than a single mixed cloud, indicating that the unified queries organize instances into coherent semantic neighborhoods (potentially reflecting fine-grained appearance modes beyond coarse class labels).
- **Continuous occlusion gradient (intra-island variation).** Within many islands, the point colors (ground-truth r_{occ}) are not randomly mixed; instead, they often show smooth transitions from low-occlusion to high-occlusion regions. For instance, the zoom-in views for Beverage (a) and Food (b) reveal distinct columnar structures where instances transition from visible (bottom) to occluded (top) along the manifold’s local axis. This suggests that occlusion is encoded as a *continuous factor* within the query representation.

These patterns provide visual evidence that the decoder queries behave as structured instance-level representations with *partial factorization*: semantic identity largely determines which island a query belongs to, while occlusion severity

modulates the position within the island. Such a structured embedding space supports the effectiveness of a unified query formulation in downstream heads that must jointly reason about visible and occluded content under diverse occlusion conditions.

A.6. Performance Visualization

Our model generalizes well across datasets with distinct image characteristics and occlusion patterns. As shown in Figure 7, it produces coherent amodal instance masks in crowded retail settings (3D-IRAIS/Real-IRAIS) and maintains stable performance on diverse everyday scenes (D2SA/COCOA-cl), indicating strong robustness to clutter and heavy occlusions.

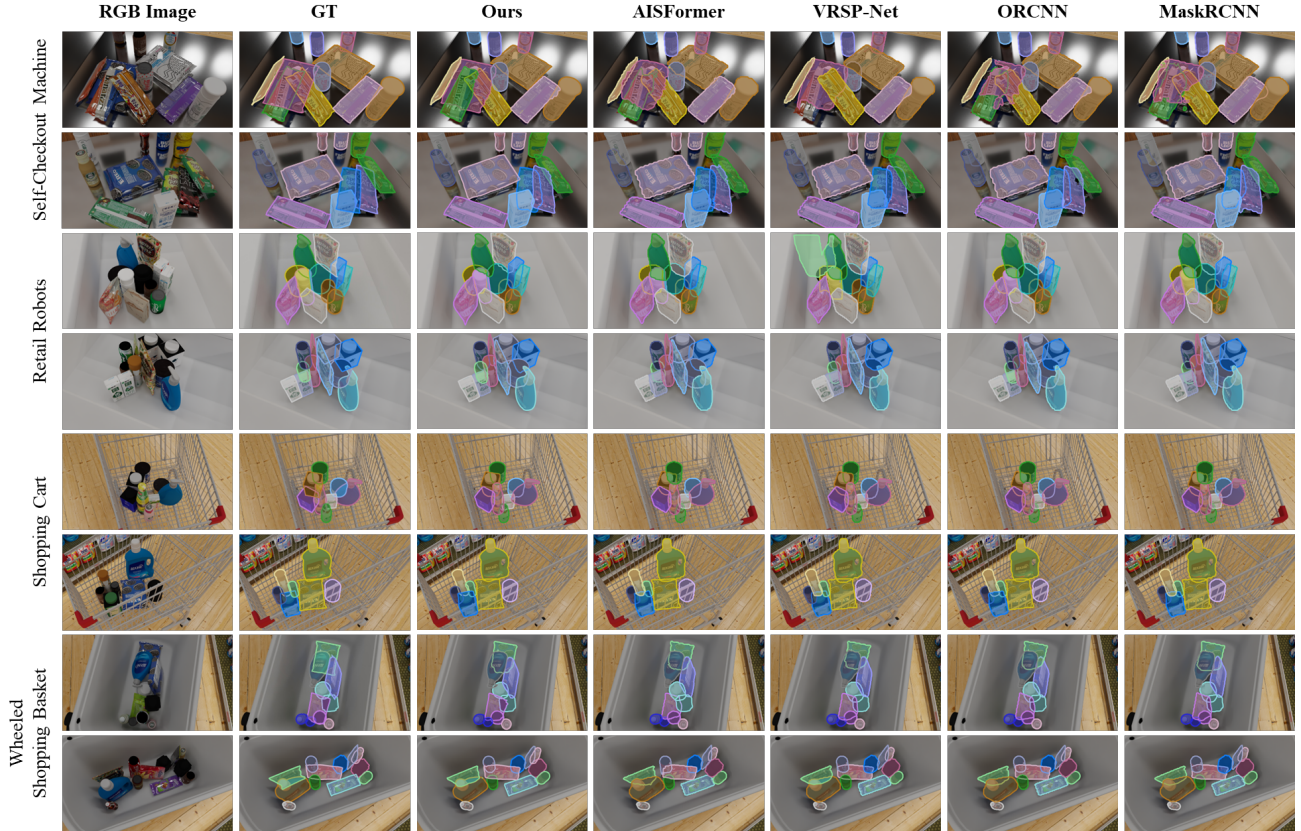


Figure 7. Qualitative results of our model on four AIS benchmarks: 3D-IRAIS, Real-IRAIS, D2SA, and COCOA-cl. Each example overlays the predicted amodal instance masks on the RGB image. The results demonstrate robust amodal completion under severe occlusions and clutter across both retail and in-the-wild scenes.

As shown in Figure 8, we provide qualitative visualizations of our model predictions to illustrate its amodal segmentation capability under diverse object scales and occlusion levels. Specifically, the examples include both small and large objects, as well as lightly and heavily occluded instances. In all cases, our model produces plausible amodal completions with smooth and coherent boundaries, while maintaining clear instance separation in cluttered scenes. These results suggest that the proposed approach is robust to scale variation and severe occlusions, and can recover complete object extents beyond the visible evidence.

B. Limitations

While our method (EUREKA) demonstrates strong performance across a wide range of common scenes, qualitative inspections suggest that it remains challenging to handle translucent/transparent objects and densely overlapping configurations. As illustrated in Figure 9, the model tends to exhibit reduced accuracy in recovering fine-grained contours and boundaries of semi-transparent objects (e.g., plastic containers in Figure 9 (a) and (c); transparent street signs in Figure 9 (g); umbrellas in Figure 9 (h)). This degradation is likely related to the inherent ambiguity of visual cues for transparent materials, where

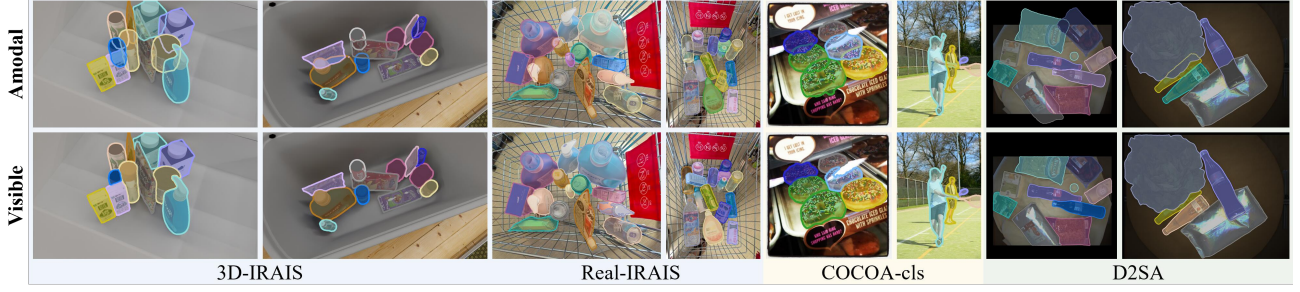


Figure 8. Qualitative visualization of our model predictions. Columns are grouped by different datasets/domains (3D-IRAIS, Real-IRAIS, COCOA-cls, and D2SA). The top row shows predicted *amodal* instance masks and the bottom row shows the corresponding *visible* masks for the same images. Our method performs consistently well across object scales (small/large) and occlusion severities (light/heavy), producing smooth boundaries and plausible amodal completions in cluttered scenes.

light refraction, background-texture blending, and dynamic occlusions can weaken appearance consistency and complicate feature extraction and correspondence estimation. Moreover, for scenes involving densely overlapped thin reflective surfaces (Figure 9 (e) and (f)), the method may occasionally produce imprecise boundary localization or imperfectly separate adjacent instances. These observations point to promising directions for future work, including integrating physics-informed priors for transparent materials and developing more robust occlusion-aware modeling to further improve generalization in such challenging cases.

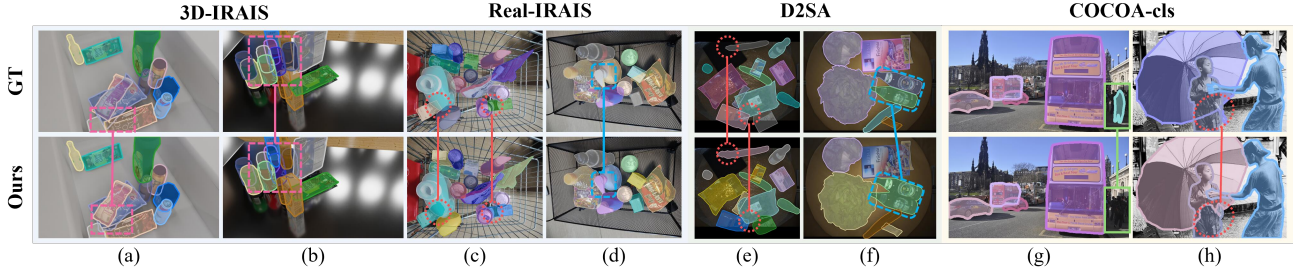


Figure 9. Qualitative limitations of our model on 3D-IRAIS, Real-IRAIS, D2SA, and COCOA-cls: ground truth amodal instance masks (top row) vs. our predictions (bottom row), showing performance gaps in handling translucent/transparent objects and densely overlapping scenes.

C. IoU “Over-Prediction” Penalty Induced by GT Definition Mismatch

To investigate the factors contributing to EUREKA’s AP_{50} performance variance in the Amodal column of Table 5, we conduct a detailed qualitative analysis of the prediction results. As visualized in Figure 10, we attribute the performance drop to three primary factors: (1) misalignment between the model’s structural priors and inconsistent ground-truth annotation standards, (2) challenges in detecting small-scale or low-contrast objects, and (3) high uncertainty in recovering the topology of non-rigid deformable objects.

Annotation Inconsistency vs. Learned Structural Priors. A significant portion of the AP_{50} penalty stems from the model’s tendency to predict “semantically complete” masks, which conflicts with “visually restricted” Ground-Truth (GT) annotations.

- **Definition Inconsistency:** As shown in Figure 10 (d) and (e), the annotation standard for categories like “Umbrella” varies; some instances include the handle in the amodal mask (e), while others exclude it (d). Driven by the unified query mechanism, EUREKA learns holistic object concepts and consistently predicts complete structures (e.g., adding the handle in (d)). This leads to an “over-prediction” penalty where the predicted mask is significantly larger than the GT, reducing the IoU below 0.5.

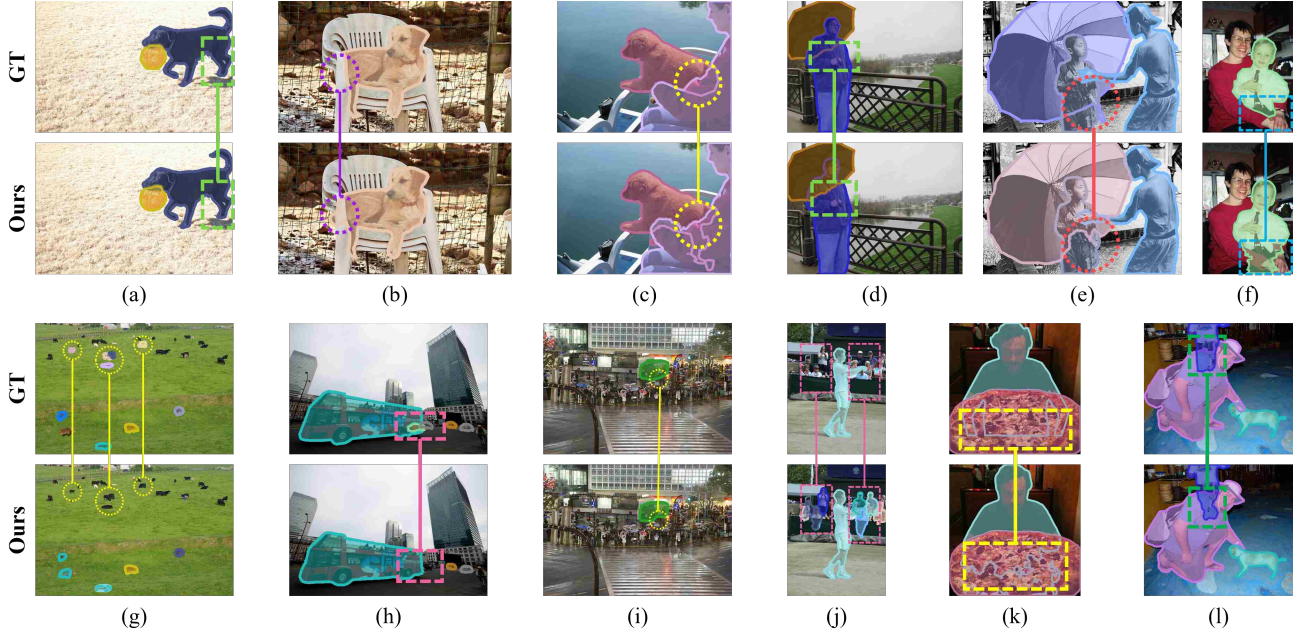


Figure 10. Qualitative analysis of failure cases on COCOA-cls. Columns compare Ground Truth (GT) and EUREKA predictions. (a)–(f) demonstrate annotation inconsistencies and ambiguous definitions of amodal extent (e.g., inconsistent umbrella handles in d vs. e). (g)–(i) highlight failures in small object detection and boundary adherence under low contrast. (j)–(l) illustrate the challenge of recovering occlusion for non-rigid “Person” categories with high pose variability.

- **Visible-as-Amodal Ambiguity:** In cases like Figure 10 (f), the GT strictly restricts the amodal mask to the visible region of the person. However, our model infers the occluded body parts based on learned priors. While semantically plausible, this prediction is penalized as a False Positive under strict IoU metrics.
- **Shape Mismatch Due to Subjectivity:** Even when occlusion is annotated, subjectivity leads to mismatches. In Figure 10 (b) and (c), the model attempts to recover the dog’s occluded hindquarters. In (c), the model infers a leg-like structure, whereas the GT follows an arbitrary curve. Such discrepancies between the model’s geometric prior and subjective annotations result in lower boundary quality scores.

Challenges with Small Objects and Low Visibility. The model exhibits limitations when handling hard examples characterized by extreme scale, low contrast, or severe occlusion.

- **Small Instances:** As seen in Figure 10 (g), the model fails to detect or accurately segment distant, tiny instances (e.g., grazing cows). This suggests that the **token-based feature representation** within the ViT backbone imposes a bottleneck; tiny objects may lack sufficient **spatial support** within the token sequence to be resolved effectively.
- **Missed Detection and Boundary Blurring:** In Figure 10 (h), severe occlusion leads to missed detections, as the vehicle heavily obstructed by the bus is not recalled by the model. Additionally, in (i), objects sharing similar colors with the background (e.g., the vehicle undercarriage blending with the road) suffer from boundary leakage. The lack of strong edge cues in these low-contrast regions hinders the mask head’s ability to delineate precise contours.

Topological Ambiguity in Deformable Objects. The “Person” category represents a unique challenge due to its non-rigid nature and high pose variability.

- **Pose Uncertainty:** Unlike rigid objects (e.g., cars) with predictable shapes, occluded human body parts can exist in numerous configurations (standing, sitting, squatting). As shown in Figure 10 (j)–(l), the model struggles to infer the correct pose for the occluded lower body. In (j), while the model correctly detects the standing player, its attempt to segment the occluded legs of the background audience results in low-quality, amorphous shapes.

- **Incomplete Recovery:** In (k) and (l), the model fails to predict the occluded lower limbs entirely. This reflects a limitation in reasoning capability: when visual cues are heavily truncated and the potential search space for the occluded topology is too vast, the model tends to output conservative or fragmented masks rather than a complete coherent shape.